Practice Set 2 Paper 4 – Recent Advances

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Practice paper compiled from the question bank.

Q1. Discuss in detail anatomy of prostate with special reference to newer concepts in surgical management of CA prostate

Type: LAQ (25 marks) (anatomy also answered in Practice Set 1 Paper 1, Q2; robotic prostatectomy in 2023-Winter Paper 1, Q3b)

Introduction

- A 20-g fibromuscular gland around the proximal urethra, from bladder neck (**base**) to urogenital diaphragm (**apex**).
- Modern RP (radical prostatectomy) is anatomical: each new technique gives a clear margin while saving a named structure (nerves, sphincter or anterior support).

Zonal anatomy (McNeal, 1981)

Zone	Share of glands	Surgical relevance
Peripheral (PZ)	About 70%	**70% of cancers**; posterolateral, so ECE (extracapsular extension) runs along the nerves
Central (CZ)	About 25%	Cone around the ejaculatory ducts; few cancers
Transition (TZ)	About 5%	BPH (benign prostatic hyperplasia); anterior cancers missed by DRE (digital rectal examination)
AFMS (anterior fibromuscular stroma)	No glands	Anterior tumours reach it; margin risk in posterior approaches

- There is **no true capsule**, so cancer breaching the gland is "extraprostatic extension".

Fascial planes and nerve sparing

Plane	Lies	Trade-off
Intrafascial	Between capsule and prostatic fascia	Most nerves; more positive margins
Interfascial	Between prostatic and levator fascia	**Standard** for organ-confined disease
Extrafascial	Outside levator and Denonvilliers fascia	Widest margin; ED (erectile dysfunction) likely

- **Denonvilliers fascia** stays on the rectum when ECE risk is low; it is excised when posterolateral ECE is suspected.

Neurovascular bundle (NVB)

- Walsh and Donker (1982). The pelvic plexus (S2–S4 parasympathetic plus hypogastric sympathetic) gives the **cavernous nerves**, running posterolaterally **3–5 mm from the capsule**, marked by capsular vessels.
- At the apex they lie at **5 and 7 o'clock**, swinging to 3 and 9 o'clock at the mid-urethra; most nerve injuries occur here.
- Nerve distribution is diffuse: a main posterolateral bundle, a posterior "hammock" and anterolateral fibres (basis of high anterior release).
- Injury: incision, entrapment, **heat**, **traction**; so dissect athermally without traction.

Vessels and lymphatics

- Inferior vesical artery (urethral and capsular branches). Preserve **accessory pudendal arteries**; they may be the main penile supply.
- DVC (dorsal venous complex): CO_2 pneumoperitoneum tamponades it, so robotic blood loss is low.
- Lymph: obturator, external and internal iliac, common iliac, presacral.

Continence anatomy

Passive	Active
Smooth muscle of bladder neck and urethra; anterior suspensory system (endopelvic fascia, detrusor apron, puboprostatic ligaments, DVC)	Horseshoe striated sphincter at the apex; pudendal nerve branches enter it at the apex

Newer concepts in surgical management

1. **RARP (robot-assisted RP):** 3D, 10× view and wristed instruments removed the suturing barrier. **Yaxley 2016** RCT (randomised controlled trial): equal function and early oncology versus open; less blood loss (300 versus 1400 mL), shorter stay. Surgeon matters more than platform.
2. **Risk-graded nerve sparing:** each side is graded from full (intrafascial) through partial to none (for example the Tewari four-grade scale). The choice uses grade, positive cores and mpMRI (multiparametric magnetic resonance imaging) capsular contact.
3. **NeuroSAFE:** intraoperative frozen section of each posterolateral surface; a positive section means resecting that NVB. It aims to allow more nerve sparing without more positive margins; the UK NeuroSAFE PROOF RCT (randomised controlled trial) tested its effect on erectile function.
4. **Anterior-support preservation:**
    - **Retzius-sparing** (Bocciardi, 2010): fully posterior via the pouch of Douglas. Earlier continence at 3 months, equal at 12; more positive margins, so avoid for anterior tumours.
    - **Hood technique** (2021): anterior approach keeping detrusor apron, puboprostatic ligaments and DVC; early data.
5. **Apical dissection:** the apex is the commonest positive-margin site. Keep maximal urethral length, watch for an apical lip, divide sharply without cautery; control the DVC distally or late. Posterior (Rocco) reconstruction aids early continence.
6. **PLND (pelvic lymph node dissection) templates:**

Template	Nodes	Drainage captured
Limited	Obturator	—
Standard	+ External iliac	About 38%
Extended	+ Internal iliac	About 63%
+ Common iliac	+ Common iliac	About 75%

- Extended template when nodal risk exceeds 5% (EAU [European Association of Urology]); it stages but does not improve survival. Lymphocele is commonest.

7. **Single-port robot (da Vinci SP, 2018):** one 2.5-cm cannula; extraperitoneal or transvesical; same-day discharge.
8. **Focal therapy anatomy:** ablate the **index lesion**. Transrectal HIFU (high-intensity focused ultrasound) suits posterior PZ lesions; anterior lesions need transperineal cryotherapy or TULSA (transurethral ultrasound ablation). Spare sphincter, NVB and rectum.

- **Also know:** PSMA PET (prostate-specific membrane antigen positron emission tomography) staging; salvage robotic RP; new platforms; ERAS (enhanced recovery after surgery).

Diagram to draw: axial prostate showing the three fasciae, the NVB at 5 and 7 o'clock and the intra-, inter- and extrafascial planes.

Key points to remember:

- PZ gives 70% of cancers; no true capsule.
- Interfascial plane is the standard nerve-sparing plane; apex is the commonest margin and nerve-injury site.
- Retzius-sparing: earlier continence, more positive margins.
- Extended PLND captures about 63% of drainage.

Q2. Describe etiopathogenesis of VUR and recent trends in management

Type: LAQ (25 marks) (VUR also answered in Practice Set 1 Paper 1, Q3f and 2020-Summer Paper 1, Q2; detailed management of bilateral VUR with surgical techniques: see Practice Set 2 Paper 3, Q2)

Definition and epidemiology

- VUR (vesicoureteral reflux) is retrograde flow of urine from the bladder into the upper tract.
- Found in about **30% of children with UTI (urinary tract infection)** and about 17% imaged for hydronephrosis. The younger the child, the likelier VUR.
- Infant boys predominate (higher voiding pressures, more UTI if uncircumcised); overall girls twice as often. Less common in Black children.

Normal antireflux mechanism

- Balance of three factors: **ureteric peristalsis**, **UVJ (ureterovesical junction) anatomy** and **bladder dynamics**.
- **Passive flap valve:** the intramural ureter is compressed against detrusor backing as the bladder fills. This needs adequate tunnel length and fixation.
- **Paquin:** tunnel length to diameter about **5:1** normally versus 1.4:1 in refluxing units. Orifice shape also matters.
- Waldeyer's sheath anchors the intramural ureter to the deep trigone.

Etiopathogenesis

1. Embryology (Mackie and Stephens)

	Early budding	Late budding
Orifice	High and lateral	Distal and medial
Result	Short tunnel, **VUR**	Ectopic, often obstructed ureter
Kidney	Abnormal bud–metanephros contact: **dysplasia**	—

- This single fault explains why high-grade VUR and dysplasia occur together.

2. Primary versus secondary

Primary	Secondary
Congenitally deficient UVJ: short tunnel, weak detrusor backing, lateral orifice	Normal UVJ overwhelmed by pressure: PUV (posterior urethral valves), neurogenic bladder (VUR in up to 30% of spina bifida), dysfunctional voiding; or UVJ anomaly (ureterocele, duplication, paraureteral diverticulum)

- They overlap: "primary" VUR often has a treatable secondary component.

3. Bladder and bowel dysfunction (BBD): high voiding pressures and incomplete emptying distort the UVJ. Detrusor overactivity is the commonest urodynamic finding. Infant boys have high voiding pressures that settle with maturation.

4. Genetics: sibling prevalence about 30%, highest under 3 years; identical twins almost fully concordant; polygenic.

5. Infection: cystitis lowers the reflux threshold; *E. coli* endotoxin causes ureteric atony.

Pathogenesis of renal damage

	Reflux-associated dysplasia	Acquired scar
Cause	Abnormal bud (congenital)	Infected urine plus VUR
Typical	Boy, grade IV–V, antenatal hydronephrosis	Girl, recurrent febrile UTI, BBD
Preventable	No	Yes

- **Intrarenal reflux** through flat compound papillae at the poles needs only 2 mmHg in neonates, so scars are polar.
- **Sterile reflux rarely scars** unless pressures are high (Ransley and Risdon). Most scars follow early infections ("big bang", now debated).
- Sequelae: hypertension in **34–38%** of scarred children; CKD (chronic kidney disease); poor growth; pre-eclampsia.

International grading (1981)

Grade	Appearance on VCUG (voiding cystourethrogram)
I	Ureter only
II	Up to calyces, no dilatation
III	Mild–moderate dilatation, minimal forniceal blunting
IV	Moderate dilatation and tortuosity
V	Gross dilatation and tortuosity; papillary impressions lost

- Natural history: grade II resolves in 63–85%, grade III about 50%; grade IV–V and bilateral III rarely.

Recent trends in management

1. **Treat infection risk, not reflux:** risk stratification by sex, age, grade, laterality, BBD, scars and presentation (Hidas model). Breakthrough UTI on prophylaxis at 2 years: 8.6% low, 26% intermediate, 62.5% high risk.
2. **Fewer VCUGs:** ultrasound first (AAP [American Academy of Pediatrics] 2011, NICE [National Institute for Health and Care Excellence]); or "top-down" DMSA (dimercaptosuccinic acid) first. Contrast-enhanced voiding urosonography avoids radiation.
3. **Objective measures:** UDR (ureteral diameter ratio) predicts resolution and breakthrough UTI better than grade; machine-learning models.
4. **Selective CAP (continuous antibiotic prophylaxis): RIVUR** halved recurrent UTI (23.6% to 12.9%) but did not reduce scarring. NNT (number needed to treat) is 5 in high-risk and 18 in low-risk children. The Swedish Reflux Trial showed benefit in girls with dilating VUR.
5. **BBD first:** timed voiding, constipation treatment, anticholinergics. Topical steroid for phimosis in boys.
6. **Endoscopic injection:** Dx/HA (dextranomer/hyaluronic acid); HIT and double HIT (hydrodistension implantation technique) give about 90% resolution. Vantris works better but can obstruct late.
7. **RALUR-EV (robot-assisted extravesical reimplantation):** less pain and shorter stay; results approaching the open benchmark of **95.9%**.
8. **Less follow-up imaging:** VCUG may be omitted after surgery for low-grade VUR if ultrasound is normal.

Diagram to draw: normal long oblique tunnel versus short lateral refluxing tunnel, with an early-budding Y-diagram.

Key points to remember:

- Paquin 5:1 normal versus 1.4:1 refluxing; early budding causes VUR and dysplasia.
- Sterile reflux rarely scars; 34–38% of scarred children become hypertensive.
- RIVUR: CAP halves UTI, not scars.
- Trend: risk-stratified care, fewer VCUGs, BBD first, endoscopic or robotic correction.

Q3a. Laparoscopic donor nephrectomy

Type: SAQ (10 marks) (living donation also in 2023-Winter Paper 1, Q3c; the pros and cons are asked again in Practice Set 2 Paper 4, Q3d)

Introduction

- **Ratner (1995)**; now the preferred approach (open under 2% in the United States). **Donor safety comes first.**
- Work-up includes measured GFR (glomerular filtration rate) and CT (computed tomography) angiography.

Side selection

- **Left preferred:** longer renal vein, simpler dissection.
- **Right** if the left has multiple arteries or complex veins, or the right is smaller or abnormal: the donor keeps the better kidney. The right vein is short.

Techniques

Technique	Features
Pure transperitoneal laparoscopic	3–4 ports; colon reflected; Pfannenstiel extraction
Hand-assisted (about two-thirds of US cases)	Hand port is also the extraction site; tactile feedback, quick haemostasis
Retroperitoneoscopic	Balloon-made space; no bowel mobilisation; direct hilum; less room
Robotic; LESS (laparoendoscopic single-site surgery)	Wristed dissection; single umbilical incision; cost

Key steps

1. Reflect colon; divide gonadal, adrenal and lumbar veins (left).
2. Keep periureteric tissue and the gonadal–ureter triangle (ureteric blood supply).
3. Hydration and diuretic before vessel division.
4. Divide ureter, artery, then vein (stapler; see Q3f on Hem-o-lok).
5. Extract quickly; cold flush.

Warm ischaemia

- Arterial occlusion to cold flush: a few minutes, slightly longer than open, without effect on long-term graft function.

Pros and cons (versus open)

Pros	Cons
Less pain, shorter stay, faster return to work	Longer operation; learning curve
Better cosmesis; **more people volunteer to donate**	Slightly longer warm ischaemia
Magnified view of hilum	Shorter vessels, especially right vein
Equal graft function and survival	Pneumoperitoneum lowers renal blood flow; cost

- **Risks:** complications about 16% (mostly gastrointestinal); mortality **3 per 10,000**; conversion 0.26–3.34%. Lifetime ESRD (end-stage renal disease) 90 per 10,000 versus 14 in matched non-donors.

Diagram to draw: left hilum with gonadal, adrenal and lumbar veins and the stapled artery and vein.

Key points to remember:

- Ratner 1995; left kidney preferred (longer vein).
- Hand-assisted in about two-thirds of US cases.
- Mortality 3 per 10,000; conversion 0.26–3.34%.
- Never rely on Hem-o-lok alone on the donor renal artery.

Q3b. Green light lasers

Type: SAQ (10 marks) (also answered as KTP laser in Practice Set 1 Paper 4, Q3e; lasers in BPH in 2017-Winter Paper 2, Q3a)

Physics

- A 1064 nm Nd:YAG (neodymium:yttrium-aluminium-garnet) beam is passed through a **KTP (potassium titanyl phosphate) or LBO (lithium triborate) crystal**. This doubles its frequency and halves the wavelength to **532 nm** (green).
- Green light is absorbed by **haemoglobin**, not water, so vascular prostate absorbs it avidly.
- Optical penetration **0.8 mm**, tissue effect 1–3 mm, thin (0.2 mm) coagulation rim. The result is bloodless **photoselective vaporisation**.

Platforms

System	Power	Note
KTP	80 W (later 100 W)	Popularised PVP (photoselective vaporisation of the prostate)
LBO HPS (High Performance System)	120 W	Faster vaporisation
LBO XPS	**180 W**	750 µm MoXy fibre: metal tip, fluid cooling

Technique

- 23–24 Fr cystoscope; side-firing fibre emitting at **70°**; saline irrigation.
- Open the bladder neck first, away from the orifices; sweep 15–30° in 1–2 seconds at ≤2 mm.
- Large bubbles mean efficient vaporisation. Pulsatile bleeding: coagulation mode (about 30 W).
- More than 5 kJ/cc predicts complete vaporisation.

Indications (AUA [American Urological Association])

- Prostates <30 cc and 30–80 cc; men at **higher bleeding risk**, who can continue anticoagulants. Not >80 cc (high retreatment).

Evidence: GOLIATH trial

- RCT (randomised controlled trial), 29 European centres, about 270 men, prostate ≤100 cc: 180 W PVP versus TURP (transurethral resection of the prostate).
- PVP **non-inferior** for IPSS (International Prostate Symptom Score), Qmax and complications to 2 years. Shorter catheter time and stay; TURP faster. Retrograde ejaculation (about two-thirds) and reoperation equal.

Complications

- Dysuria (highest of transurethral techniques); UTI (urinary tract infection) and epididymitis from coagulum; capsular perforation 0.2–1%.
- No TUR (transurethral resection) syndrome; transfusion very rare; **no tissue for histology**.

Recent advances

- GreenLight enucleation; office and same-day catheter-free pathways.

Diagram to draw: side-firing fibre at 70° vaporising a lateral lobe.

Key points to remember:

- 532 nm, frequency-doubled Nd:YAG; haemoglobin chromophore; 0.8 mm penetration.
- 180 W XPS with MoXy fibre.
- GOLIATH: PVP non-inferior to TURP.
- Best for anticoagulated men; not >80 cc.

Q3c. Single port lap. Surgeries in urology

Type: SAQ (10 marks) (also answered as a long answer in 2023-Winter Paper 4, Q1)

Read as single-port (single-site) laparoscopic and robotic surgery in urology.

Definition

- LESS (laparoendoscopic single-site surgery): all instruments and the camera pass through **one skin incision**, usually umbilical. The term was agreed by LESSCAR (LESS Consortium for Assessment and Research, 2008).
- Variants: pure LESS; "LESS + 1" (extra 2–5-mm port); R-LESS (robotic); purpose-built **SP (single-port) robot**.
- Aim: less pain, better cosmesis, shorter stay.

Access

- Several low-profile trocars in one fascial incision, or **multichannel ports** (TriPort, GelPOINT).
- **da Vinci SP (FDA [Food and Drug Administration] approval 2018):** one 25–27-mm cannula carrying three wristed instruments and an articulating camera. Needs 10–25 cm from cannula to target.

Problem: lost triangulation

- Instruments lie parallel to the camera: clashing, poor depth perception, hard suturing.
- **Solutions:** articulating or pre-bent instruments crossing at the fascia (chiral manipulation); flexible-tip scopes; instruments of different lengths; an extra needlescopic port; robotic platforms.

Indications

- **Kidney:** simple, donor, radical and partial nephrectomy; pyeloplasty; nephroureterectomy; cyst decortication.
- Adrenalectomy; varicocelectomy; paediatric orchidopexy and pyeloplasty.
- **Pelvis:** SP radical prostatectomy (extraperitoneal or transperitoneal); transvesical simple prostatectomy; diverticulectomy.
- Best for thin patients and small or benign lesions. Avoid in obesity, large or hilar tumours and dense adhesions.

Outcomes versus conventional laparoscopy

- RCT (randomised controlled trial): same time, blood loss, complications and stay; **less analgesia**.
- Donor nephrectomy meta-analysis: lower pain scores, otherwise equal.
- SP robotic partial nephrectomy: shorter stay, less opioid; **longer warm ischaemia** in one review.
- An extra port is often needed. Gains are limited to pain, cosmesis and stay.

Complications

- **Incisional hernia** (larger fascial defect); conversion; longer operation; surgeon strain; usual laparoscopic injuries.

Recent advances

- SP retroperitoneal renal surgery; same-day discharge; magnetic anchoring instruments.

Diagram to draw: umbilical multichannel port with crossed articulating instruments versus a triangulated multiport layout.

Key points to remember:

- LESSCAR 2008 standardised the term LESS.
- Core problem: lost triangulation.
- Same outcomes as laparoscopy; less pain; more hernias.
- da Vinci SP (2018): one 25–27-mm cannula.

Q3d. Pros and cons of lap donor nephrectomy

Type: SAQ (10 marks)

See Practice Set 2 Paper 4, Q3a (full answer).

Key points to remember (pros and cons angle):

- Pros: less pain, shorter stay, earlier return to work, better cosmesis, and more willing donors; graft function equals open.
- Cons: longer operation, learning curve, slightly longer warm ischaemia, shorter vessels (especially on the right), cost.
- Safety: mortality 3 per 10,000; conversion 0.26–3.34%.
- Do not use Hem-o-lok clips alone on the donor renal artery.

Q3e. Holmium laser

Type: SAQ (10 marks) (holmium enucleation in detail in 2023-Winter Paper 4, Q2; stone energy sources in Practice Set 1 Paper 4, Q2)

Physics

- Ho:YAG (holmium:yttrium-aluminium-garnet), a solid-state, flashlamp-pumped, **pulsed** laser at **2120 nm** (often quoted as 2100 nm).
- Chromophore **water**; penetration about **0.4 mm**. It cuts and vaporises; heat spreads 2–3 mm and seals small vessels.
- Flexible quartz fibres; works in saline.

Mechanism on stones

- Long pulses (>250 µs) act **photothermally**, not by shock waves: the stone melts or decomposes.
- **Moses effect:** the start of each pulse makes a vapour channel; the rest of the pulse crosses it to the stone. So keep the fibre close and perpendicular.
- Breaks **all stone types**, including cystine and calcium oxalate monohydrate.

Settings

Mode	Settings	Effect
Fragmenting	0.6–1.0 J × 6–10 Hz	Pieces for basketing; more retropulsion
Dusting	0.2–0.3 J × 40–80 Hz	Fine dust; "popcorning" in a calyx
HoLEP enucleation	2.0 J × 40–50 Hz	Cutting along the capsule

- Retropulsion rises with pulse energy, falls with longer pulses or pulse modulation.
- Heat tracks power: ≤10 W in the ureter; good outflow.

Uses

- **Stones:** URS (ureteroscopy), RIRS (retrograde intrarenal surgery), PCNL (percutaneous nephrolithotomy), bladder stones.
- **BPH (benign prostatic hyperplasia):** HoLEP (holmium laser enucleation of the prostate): size-independent, the only endoscopic procedure superior to TURP (transurethral resection of the prostate), safe on anticoagulants.
- **Incisions:** strictures, bladder neck contracture, endopyelotomy, ureterocele.
- **Tumours:** UTUC (upper tract urothelial carcinoma) ablation; en bloc bladder tumour resection.

Safety

- Eyewear for all; thermal urothelial injury; steep HoLEP learning curve.

Recent advances

- High-power (120 W) pulse-modulated machines. TFL (thulium fibre laser, 1940 nm) beat low-power holmium but equalled high-power holmium in RCTs (randomised controlled trials).

Diagram to draw: vapour bubble (Moses channel) between fibre tip and stone.

Key points to remember:

- 2120 nm; water chromophore; pulsed; 0.4 mm penetration.
- Photothermal; Moses effect: fibre close and perpendicular.
- Fragment 0.6–1.0 J × 6–10 Hz; dust 0.2–0.3 J × 40–80 Hz; ≤10 W in ureter.
- HoLEP: size-independent, superior to TURP.

Q3f. Hemostasis in lap surgery

Type: SAQ (10 marks) (haemostasis in laparoscopic partial nephrectomy in 2019-Summer Paper 3, Q1; energy sources in 2015-Summer Paper 2, Q3d)

Principles

- Devices seal by **denaturing protein**. Plan from imaging; keep a haemorrhage tray ready.

Effect of pneumoperitoneum

- At 12–15 mmHg CO_2 **tamponades venous bleeding**, hiding it.
- Raise to 20–25 mmHg briefly for venous bleeding. Before exit, **inspect at low pressure (5–10 mmHg)** to unmask hidden bleeders and port-site bleeding.

Energy devices

Device	Lateral spread	Note
Monopolar	5–7 mm	Capacitive coupling, insulation failure
Bipolar	1–6 mm	Current only between jaws
Vessel sealer (LigaSure)	1–3 mm	Seals vessels **≤7 mm**; one application
Ultrasonic (Harmonic)	≈1 mm	Blade >200 °C; cools slowly; no current
Argon beam	Superficial	Raw surfaces; gas embolism

Clips

- Titanium clips: no diathermy near them (slip).
- **Hem-o-lok** (polymer, locking): circumferential dissection, see the tip, feel the snap, at least two clips on the patient side with a 2-mm cuff.
- **FDA (Food and Drug Administration) warning:** after donor renal artery stump haemorrhage, the manufacturer contraindicated Hem-o-lok on the **living-donor renal artery**. A later series found it safe with correct technique, but use a stapler there.
- Lapra-Ty absorbable clips act as knots.

Staplers

- Endo GIA through a 12-mm port; **grey 2.0 mm or white 2.5 mm** for renal vessels; blue or green for thicker tissue.
- **Never fire across clips** (misfire). Reinforce a bleeding staple line with clips.

Sutures and haemostatic agents

- Intracorporeal or barbed sutures. Adjuncts: fibrin sealant, gelatin–thrombin matrix, oxidised cellulose.

Managing a venous injury

- Raise pressure → suction → pledget tamponade → sealant → extra port for suturing or Satinsky → hand-assist → convert. Keep the patient well filled (gas embolism).
- Arterial: minor, tamponade; renal artery, staple; aortic, convert.

Diagram to draw: Hem-o-lok placement with two patient-side clips and a 2-mm cuff.

Key points to remember:

- Pneumoperitoneum hides venous bleeding: inspect at low pressure before exit.
- LigaSure seals ≤7 mm; spread monopolar 5–7 mm, ultrasonic ≈1 mm.
- Hem-o-lok contraindicated on the donor renal artery.
- Never staple across clips; grey or white loads for vessels.

Rapid Revision Sheet – Practice Set 2 Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss in detail anatomy of prostate with special reference to newer concepts in surgical management of CA prostate (LAQ)

- PZ gives 70% of cancers; no true capsule.
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- Paquin 5:1 normal versus 1.4:1 refluxing; early budding causes VUR and dysplasia.
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- Trend: risk-stratified care, fewer VCUGs, BBD first, endoscopic or robotic correction.

Q3a. Laparoscopic donor nephrectomy (SAQ)

- Ratner 1995; left kidney preferred (longer vein).
- Hand-assisted in about two-thirds of US cases.
- Mortality 3 per 10,000; conversion 0.26–3.34%.
- Never rely on Hem-o-lok alone on the donor renal artery.

Q3b. Green light lasers (SAQ)

- 532 nm, frequency-doubled Nd:YAG; haemoglobin chromophore; 0.8 mm penetration.
- 180 W XPS with MoXy fibre.
- GOLIATH: PVP non-inferior to TURP.
- Best for anticoagulated men; not >80 cc.

Q3c. Single port lap. Surgeries in urology (SAQ)

- LESSCAR 2008 standardised the term LESS.
- Core problem: lost triangulation.
- Same outcomes as laparoscopy; less pain; more hernias.
- da Vinci SP (2018): one 25–27-mm cannula.

Q3d. Pros and cons of lap donor nephrectomy (SAQ)

- Pros: less pain, shorter stay, earlier return to work, better cosmesis, and more willing donors; graft function equals open.
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- 2120 nm; water chromophore; pulsed; 0.4 mm penetration.
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